



Energy Economics

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Does green finance really inhibit extreme hypocritical ESG risk? A greenwashing perspective exploration

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Highlights

- The development of green finance can alleviate firm's greenwashing.
- Green finance works through increasing firms' green innovation, profitability, and reducing financial constraint.
- The improvement effect of green finance on greenwashing is more pronounced in state -owned firms.
- The improvement effect of green finance is more pronounced in pollution-intensive and highly regulated firms.

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Abstract

This paper investigates the link between green finance development and firms' environmental, social, and governance (ESG) performance quality based on a firm-level panel dataset from 2014 to 2020. The risk of extremely hypocritical ESG disclosures greatly matters to sustainable growth. Empirical evidence indicates that the

development of green finance can play an effective role in motivating ESG performance by mitigating firms' reactions in alleviating greenwashing. We find evidence supporting moderating effects of green finance on increasing firms' green innovation activities, profitability, and reducing financial constraints level. In addition, the improvement effect of green finance on corporate greenwashing performance is more pronounced in state-owned firms, pollution-intensive firms, and firms with a higher level of environmental regulation than publicly listed corporations. Overall, this paper provides a timely assessment of the social value of green finance in emerging markets with a novel connection to ESG.

Keywords

Green finance; ESG; Greenwashing; Green innovation; Financial constraints

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Introduction

The resource dilemma and environmental pollution, represented by greenhouse gas emissions, global warming, and resource depletion, are urgently needed to be solved worldwide (Chiu and Lee, 2020; Lee and He, 2022; Su et al., 2022). In the process of economic transformation, economies have entered a stage of high-quality development. A series of issues are closely related to sustainable development, including the carbon neutrality goal, energy conservation, environmental protection, rejuvenation of the country through science and technology, and common prosperity (e.g., Gu et al., 2021; Li et al., 2021). These sustainable requirements may impact corporate decision-making and behavior (Ren et al., 2022). As a result, corporate managers, financial institutions, and policymakers increasingly focus on firms' environmental, social, and governance (ESG) information (Deng et al., 2023; Zhang et al., 2021; Zhang and Wong, 2022). ESG has not only become a crucial standard for evaluating the comprehensive level of firms but also a key tool for implementing the carbon neutrality strategy and promoting green finance, which has earned great attention and wide concern from the whole society (Wu and Huang, 2022).

According to the definition discussed in previous literature, ESG is not limited to ESG ratings or scores, but ESG practices, policies and information disclosure are also extensively involved in the concept (Friede et al., 2015). Under the impetus and frameworks of ESG, the essence of sustainable development theory indicates that firms pay more attention to environmental, social responsibility, and corporate governance issues (Christensen et al., 2021). The economic externality theory further demonstrates that ESG practice produces spillover effects through the collection of pollution discharge fees, the establishment of carbon emission trading rights, and subsidies for new energy vehicles (Eliwa et al., 2021; Hao and He, 2022; El Khoury et al., 2022). Moreover, ESG advocates consider the rights and interests of stakeholders, which is in line with the requirements of corporate social responsibility theory (Capelli et al., 2021). Solving ESG issues has become a focus of risk management for investors, shareholders, governments, and other stakeholders (Capelle-Blancard et al., 2019). Moreover, ESG has evolved into a new competitive strategy for firms, which affects firms' decision-making (Galbreath, 2013) and helps enterprises cultivate a good reputation (Flammer and Bansal, 2017). ESG not only attracts investors and alleviates financing constraints (Ng and Rezaee, 2015), it also improves transparency between corporate managers and capital market investors, thereby affecting capital market returns (Miralles et al., 2019).

Nevertheless, the risk of extremely hypocritical ESG efforts also exists. Selected ESG information disclosures, such as greenwashing, may have a negative impact due to artificial distortions (Zhang, 2022). Accurate ESG ratings provide investors with more information, which is beneficial in alleviating information asymmetry (Yu et al., 2020). However, external lenders and consumers may be misled by differences in ESG ratings, thereby affecting corporate decision-making and market returns in turn. Existing evidence illustrates that firms are more likely to gain lower costs for building a better reputation and developing better social image through greenwashing behaviors, according to the close relationship between ESG ratings and external lenders (Ghisetti and Rennings, 2014; Yu et al., 2020). In specific, greenwashing which hypocritically creates better ESG disclosure can be used as an alternative business strategy to gain the benefit of ESG through misleading the external rating institutions and creating information asymmetry (Zhang, 2022).

Differing from traditional finance, green finance focuses on environmental interests (Zhou et al., 2020). As a financial instrument,

green finance has been widely investigated in China and worldwide, playing a vital role in guaranteeing the sustainable development and environmental targets of the economy (Akomea-Frimpong et al., 2022). Green finance aims to provide financial support for eco-friendly projects with environmental protection as its principal goal, including financing resources, carbon finance, and operating funding (Zhou et al., 2020). In light of its working mechanisms, green finance promotes ecological sustainability by fighting for energy efficiency and against climate change (Freebairn, 2012). Green financing works by motivating capital flows toward environmentally sustainable and cleaner production activities to ramp up green production (Xu and Li, 2020). Furthermore, green finance seeks to guide industrial firms in using less energy and more clean energy capability to manage financial resources, which eventually results in obtaining sustainable economic and environmental achievements (Gu et al., 2021). In addition, previous evidence holds the view that the green finance can significantly stimulate investment in innovative technologies in certain fields, such as renewable energy (Jones, 2015), because green finance helps alleviate financial constraints and solves information asymmetry problems in green innovation (Zhang et al., 2022).

Although green finance has been increasingly prominent for transition and emerging economies, it is still worth investigating the extent to which green finance facilitates high-quality economic development. Therefore, this study sets out to explore whether green finance can be beneficial as a driver of sustainability growth, thereby curbing the risk of extreme hypocritical ESG. In the context of a high-quality and sustainable development mode, China's ESG practice is thriving, which helps provide a fruitful test environment with rich firm-level information.

In this study, we construct a Chinese listed-firm-level panel dataset covering 2014 to 2020, which contains rich information and data variables relating to ESG and fundamental firm-level characteristics for the observation period. In particular, we explore how green finance impacts ESG efficiency measured by greenwashing behavior and the probable mechanisms involved. Several findings are accordingly presented in this paper. First, we find that green finance has a significant and negative impact on greenwashing for full samples. While state-owned firms benefit significantly from green finance, foreign-owned firms do not. Heavily polluting firms strongly reduce greenwashing behaviors compared to firms with cleaner production. Moreover, firms that are highly environmentally regulated

benefit more from green finance. Second, regarding the exploration of potential mechanisms, environmentally oriented innovation activities, including green innovation input and output, are significantly driven by green finance, which results in reducing the greenwashing level. Third, green finance further improves ESG performance by alleviating financial constraints, and green finance also improves profitability and ESG-related financing support, thereby decreasing greenwashing behaviors.

Based on a discussion of the existing literature, this paper can further deepen the related research in the following three aspects. First, the creation of an appropriate indicator classification for ESG performance efficiency is still in a tentative phase (Yu et al., 2020; Zhang, 2022a). Most existing studies select a single indicator to measure ESG, such as ESG rating score or disclosure score, which might be single-sided and biased, and cannot perfectly reflect the richness of ESG dimensions and the true level of ESG efficiency. We construct a greenwashing indicator measuring the gap between ESG disclosure score and ESG rating score, which well estimates ESG efficiency. Second, the potential mechanism or coordinated expansion between green finance and greenwashing is worthy of investigation.

In the context of the dynamic development of ESG performance, the mechanisms of green finance on green innovation include not only direct but also deepening and intermediary effects (Zhang, 2022b; Zhang et al., 2022). Third, a range of signs exist showing that green finance can decrease greenwashing by alleviating financial constraints and increasing profitability. However, the causal effect between green finance and greenwashing is uncertain, and the concern of whether these mechanisms have moderating effects has not been fully addressed (Muganyi et al., 2021; Yang et al., 2021). Therefore, this paper clarifies the impact of green finance on greenwashing at different levels based on the construction of a comprehensive greenwashing indicator system. The intermediate effects are combined to empirically demonstrate the role of green finance on green innovation, which is another unique contribution and value of the existing literature.

The remainder of this article is presented as follows. Section 2 illustrates the literature on a variety of understandings of green finance and ESG performance. In Section 3, we explain the issues regarding our empirical framework, which is used identify the causal effect of green finance on greenwashing, as well as our description of

the statistical methods. Section 4 introduces a summarizing overview of our empirical results of green finance in greenwashing, along with an exploration of several mechanisms. Section 5 offers several robust empirical results and accordingly provides relevant discussions. A concluding section summarizes the main findings with policy implications.

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ESG performance and greenwashing

The positive relationship between firms' ESG information disclosure and environmental performance has been verified in existing studies (Brammer et al., 2006). ESG information disclosure can encourage firms to reduce emissions and assume social responsibilities, which suppresses the discharge of wastewater and sulfur dioxide (Chen et al., 2018), reducing greenhouse gas emissions (Downar et al., 2021). Christensen et al. (2022) further provide evidence that ESG information disclosure can ...

Data source, empirical model design, and variables

We seek to enrich the emerging ESG literature by investigating if green finance can help to curb the greenwashing risk reactions to adverse hypocritical ESG behavior. ...

Baseline regression results and discussion

Following the construction of the model shown in Eq. (1), our empirical procedure seeks to extend the emerging literature on ESG performance by examining if green finance can help to mitigate the negative reaction of greenwashing. Table 3 provides estimation results for the effect of green finance on firms' greenwashing. To make our fundamental results robust, we gradually add fixed effects and control variables in our regression.

Column 1 presents the coarse estimation result of green finance ...

Green innovation input and output mechanisms exploration

Green innovation directly affects the impact of ESG performance, as corporate ESG is significantly impacted by green innovation activities. We use two dimensions, green innovation input, and output, to identify the mediate effects of green innovation on the relationship between green finance and greenwashing.

Table 7 presents the intermediate effect estimation results on green innovation input mechanism exploration. Columns 1–2 show that green finance significantly facilitates green innovation ...

Conclusions and policy implications

ESG plays a vital role for society in sustainable and high-quality development. The advantages of ESG are widely known by both corporate managers and policymakers – understanding how to curb extreme hypocritical ESG risk and improve ESG rating system performance is vital. To contribute to ESG development, not only the quantity of information disclosure should be considered but also the quality of ESG disclosure and the completion of ESG practice should be more focused. In this way, the symbolic ...

CRedit authorship contribution statement

Dongyang Zhang: Conceptualization, Methodology, Writing – review & editing, Formal analysis, Data curation. ...

Acknowledgments

This paper acknowledges the financial support of the National Natural Science Foundation of China (72103147) and the Project of Cultivation

for Young top-notch Talents of Beijing Municipal Institutions
(BPHR202203184). ...

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